Unit: 6 Unit 8: Quadratics **Name:** ______________
Topic: Converting Forms (Completing the Square) **Date:** __________

Advanced (Carolina Reaper)

Q1. Convert the equation $x^2 + 6x + 8$ to vertex form.
(A) $(x + 3)^2 - 1$
(B) $(x + 3)^2 + 1$
(C) $(x + 4)^2 - 8$
(D) $(x + 2)^2 + 4$
(E) $(x + 1)^2 - 7$

Q2. Complete the square to transform $x^2 + 4x$ into vertex form.
(A) $x^2 + 4x + 4$
(B) $(x + 2)^2$
(C) $(x + 4)^2 - 16$
(D) $(x - 2)^2 + 4$
(E) $x^2 + 4x - 4$

Q3. What is the vertex of the parabola $y = (x - 2)^2 + 1$?
(A) (0, 1)
(B) (1, 0)
(C) (2, 1)
(D) (-1, 0)
(E) (0, 0)

Q4. Tom is planning a road trip. The cost of the trip is modeled by $y = x^2 + 5x + 6$, where y is the cost and x is the number of people. Convert the equation to vertex form.
(A) $y = (x + 3)^2 - 3$
(B) $y = (x + 2)^2 + 2$
(C) $y = (x + 4)^2 - 10$
(D) $y = (x - 1)^2 + 7$
(E) $y = (x - 3)^2 - 3$

Q5. Samantha has been saving money to buy a new bike. The amount she has saved is given by $y = x^2 + 3x - 2$, where y is the amount saved and x is the number of weeks. Complete the square to transform the equation into vertex form.
(A) $y = (x + 2)^2 - 6$
(B) $y = (x + 1)^2 - 3$
(C) $y = (x - 1)^2 + 3$
(D) $y = (x + 3)^2 - 11$
(E) $y = (x - 2)^2 + 2$

Q6. The height of a projectile is given by $y = x^2 - 7x + 10$. Convert the equation to vertex form.
(A) $y = (x - 4)^2 - 6$
(B) $y = (x - 3)^2 + 1$
(C) $y = (x + 3)^2 - 19$
(D) $y = (x + 4)^2 - 6$
(E) $y = (x - 2)^2 - 6$

Q7. A water tank can be filled at a rate modeled by $y = x^2 + 2x - 3$, where y is the amount of water and x is the time in hours. Complete the square to transform the equation into vertex form.
(A) $y = (x + 1)^2 - 4$
(B) $y = (x - 1)^2 + 2$
(C) $y = (x + 2)^2 - 7$
(D) $y = (x + 3)^2 - 12$
(E) $y = (x - 2)^2 + 1$

Q8. The distance between two cities is given by $y = x^2 - 9x + 20$. Convert the equation to vertex form.
(A) $y = (x - 5)^2 - 5$
(B) $y = (x + 4)^2 - 16$
(C) $y = (x - 4)^2 + 4$
(D) $y = (x + 5)^2 - 25$
(E) $y = (x - 3)^2 - 1$

Open Ended Questions (FRQ)

Q9. A bakery sells a total of x loaves of bread per day. The profit is given by $y = x^2 + 10x - 50$. Complete the square to transform the equation into vertex form and interpret the result. Show all steps and provide the vertex form equation.

Q10. The cost of producing x units of a product is given by $y = x^2 + 5x + 10$. Convert the equation to vertex form and find the minimum cost. Provide the minimum cost and the number of units produced at that cost. Show all steps and provide the vertex form equation.

Chef's Tips

- Tip 1: Ensure students understand the concept of completing the square.
- Tip 2: Emphasize the importance of interpreting transformed equations.
- Tip 3: Encourage students to check their work by plugging the vertex form back into the original equation.